

```
>> dynare mcmc.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 25 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
7 block(s) found:
  5 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
                                and 16 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
                                and 16 feedback variable(s).

Preprocessing completed.
Preprocessing time: 0h00m00s.
```

STEADY-STATE RESULTS:

q	29.3325
k	1.37708
R	1.01516
Rp	1.0121
b	39.7902
x	0.0126829
varphi	0.0303425
mu	0.00522154
kp	2.76292
chi	1.02332
nu	0.00416305
eta	1.38229
zeta	1.02332
N	10.8475
phi	0.272618
Ne	10.7896
Nn	0.0578853
xp	2.25694
bp	28.9427
Y	2.26962
eb	0
eN	0
eY	0
N_obs	0
Y_obs	0

EIGENVALUES:

Modulus	Real	Imaginary
0	0	0
2.719e-18	-2.719e-18	0
2.508e-17	-2.508e-17	0

9.742e-15	-9.742e-15	0
9.931e-15	5.251e-15	8.43e-15
9.931e-15	5.251e-15	-8.43e-15
2.376e-12	-2.376e-12	0
2.379e-12	2.379e-12	0
0.01476	0.01476	0
0.8	0.8	0
0.8	0.8	0
0.9878	0.9878	0
1.016	1.016	0
1.02	1.02	0
1.02	1.02	0
1.021	1.021	0
1.112e+16	1.112e+16	0
7.385e+19	7.385e+19	0
1.229e+32	-1.229e+32	0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.

===== Identification Analysis =====

Testing prior mean

Note that differences in the criteria could be due to numerical settings,
numerical errors or the method used to find problematic parameter sets.

Settings:

Derivation mode for Jacobians:	Analytic using sylvester equations
Method to find problematic parameters:	Nullspace and multicorrelation

coefficients

Normalize Jacobians:	Yes
Tolerance level for rank computations:	robust
Tolerance level for selecting nonzero columns:	1e-08
Tolerance level for selecting nonzero singular values:	1e-03

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution
matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):

All parameters are identified in the Jacobian of steady state and minimal system
(rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):

All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with
tol = robust).

MOMENTS (ISKREV, 2010):

All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol
= robust).

==== Identification analysis completed ====

You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -168787.1848

```

f at the beginning of new iteration,      168787.1847882181
Predicted improvement: 53024395774.718353271
lambda =      1; f =      10742391.6758482
lambda =    0.33333; f =      1337433.3654417
lambda =    0.11111; f =      297158.9063756
lambda =    0.037037; f =      182660.9184760
lambda =    0.012346; f =      170205.0942406
lambda =    0.0041152; f =      168907.1944704
lambda =    0.0013717; f =      168791.4070376
lambda = 0.00045725; f =          57.9683548
lambda = 0.00015242; f =          388.8403719
lambda = 5.0805e-05; f =          2260.5619304
lambda = 1.6935e-05; f =          7631.7616599
lambda = 5.645e-06; f =          25339.8165696
lambda = 1.8817e-06; f =          68303.0451293
Norm of dx      3256.5
----
Improvement on iteration 1 =      168729.216433402
-----
f at the beginning of new iteration,          57.9683548161
Predicted improvement:          12.642711738
lambda =      1; f =          54.1278601
lambda =    0.33333; f =          53.1046882
Norm of dx    0.050285
----
Improvement on iteration 2 =          4.863666623
-----
f at the beginning of new iteration,          53.1046881930
Predicted improvement:          2.256143497
lambda =      1; f =          49.0913511
lambda =    1.9332; f =          46.0703253
lambda =    3.7372; f =          41.5714185
Norm of dx    0.023526
----
Improvement on iteration 3 =          11.533269739
-----
f at the beginning of new iteration,          41.5714184541
Predicted improvement:          5.909261842
lambda =      1; f =          31.6820746
lambda =    1.9332; f =          27.0079122
Norm of dx    0.10542
----
Improvement on iteration 4 =          14.563506284
-----
f at the beginning of new iteration,          27.0079121705
Predicted improvement:          1.215009045
lambda =      1; f =          25.0222959
lambda =    1.9332; f =          23.8127099
Norm of dx    0.048101
----
Improvement on iteration 5 =          3.195202291
-----
f at the beginning of new iteration,          23.8127098795
Predicted improvement:          1.304183607
lambda =      1; f =          22.7902990
Norm of dx    0.13201
----
Improvement on iteration 6 =          1.022410922

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-----
f at the beginning of new iteration,          22.7902989576
Predicted improvement:          0.725844234
lambda =          1; f =          22.1324789
Norm of dx    0.031634
----
Improvement on iteration 7 =          0.657820021
-----
f at the beginning of new iteration,          22.1324789370
Predicted improvement:          0.014347932
lambda =          1; f =          22.1145858
Norm of dx    0.014729
----
Improvement on iteration 8 =          0.017893117
-----
f at the beginning of new iteration,          22.1145858205
Predicted improvement:          0.001204110
lambda =          1; f =          22.1133445
Norm of dx    0.004889
----
Improvement on iteration 9 =          0.001241364
-----
f at the beginning of new iteration,          22.1133444565
Predicted improvement:          0.000041874
lambda =          1; f =          22.1132625
lambda =      1.9332; f =          22.1131896
lambda =      3.7372; f =          22.1130580
lambda =      7.2247; f =          22.1128389
lambda =     13.967; f =          22.1125473
Norm of dx 0.00020527
----
Improvement on iteration 10 =          0.000797191
-----
f at the beginning of new iteration,          22.1125472650
Predicted improvement:          0.000913976
lambda =          1; f =          22.1108492
lambda =      1.9332; f =          22.1094999
lambda =      3.7372; f =          22.1075295
lambda =      7.2247; f =          22.1060725
Norm of dx 0.0017119
----
Improvement on iteration 11 =          0.006474763
-----
f at the beginning of new iteration,          22.1060725018
Predicted improvement:          0.005345432
lambda =          1; f =          22.0953945
lambda =      1.9332; f =          22.0854537
lambda =      3.7372; f =          22.0663010
lambda =      7.2247; f =          22.0295155
lambda =     13.967; f =          21.9592917
lambda =         27; f =          21.8267981
lambda =     52.196; f =          21.5824223
lambda =     100.9; f =          21.1510249
lambda =     195.07; f =          20.4517003
lambda =     377.1; f =          19.4949282
Norm of dx 0.0038532
----
Improvement on iteration 12 =          2.611144256

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-----
f at the beginning of new iteration,      19.4949282456
Predicted improvement:      2.800775392
lambda =      1; f =      13.9114453
lambda =      1.9332; f =      8.4254047
lambda =      3.7372; f =      -4.2774094
lambda =      7.2247; f =      159.0072009
lambda =      4.8647; f =      -14.1169781
lambda =      6.1676; f =      -20.7921757
lambda =      7.8194; f =      19.4950273
lambda =      6.7817; f =      5.1699723
Norm of dx      2.6216
----
Improvement on iteration 13 =      40.287103979
warning: possible inaccuracy in H matrix
-----
f at the beginning of new iteration,      -20.7921757332
Predicted improvement:      15.138021177
lambda =      1; f =      1.3047704
lambda =      0.33333; f =      -16.2990839
lambda =      0.11111; f =      -21.4487165
lambda =      0.037037; f =      -21.5386527
Norm of dx      9.1597
----
Improvement on iteration 14 =      0.746476940
-----
f at the beginning of new iteration,      -21.5386526733
Predicted improvement:      2.493641275
lambda =      1; f =      -26.2923976
lambda =      1.9332; f =      -30.3077119
lambda =      3.7372; f =      -36.6245237
lambda =      7.2247; f =      -31.6314266
lambda =      4.8647; f =      -38.9455028
lambda =      6.1676; f =      -38.2085645
Norm of dx      0.25824
----
Improvement on iteration 15 =      17.406850114
-----
f at the beginning of new iteration,      -38.9455027871
Predicted improvement:      2.228120739
lambda =      1; f =      -37.4601044
lambda =      0.33333; f =      -39.9559743
Norm of dx      0.34933
----
Improvement on iteration 16 =      1.010471497
-----
f at the beginning of new iteration,      -39.9559742839
Predicted improvement:      1.290485854
lambda =      1; f =      -41.1935199
Norm of dx      0.66842
----
Improvement on iteration 17 =      1.237545620
-----
f at the beginning of new iteration,      -41.1935199040
Predicted improvement:      0.076163203
lambda =      1; f =      -41.2419193
Norm of dx      0.11
-----

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Improvement on iteration 18 =          0.048399386
-----
f at the beginning of new iteration,      -41.2419192895
Predicted improvement:          0.006936443
lambda =          1; f =          -41.2492337
Norm of dx    0.035251
----
Improvement on iteration 19 =          0.007314374
-----
f at the beginning of new iteration,      -41.2492336635
Predicted improvement:          0.000101343
lambda =          1; f =          -41.2493785
lambda =      1.9332; f =          -41.2494105
Norm of dx    0.0013273
----
Improvement on iteration 20 =          0.000176879
-----
f at the beginning of new iteration,      -41.2494105429
Predicted improvement:          0.000097906
lambda =          1; f =          -41.2496005
lambda =      1.9332; f =          -41.2497665
lambda =      3.7372; f =          -41.2500568
lambda =      7.2247; f =          -41.2505037
lambda =     13.967; f =          -41.2509434
Norm of dx    0.00039551
----
Improvement on iteration 21 =          0.001532808
-----
f at the beginning of new iteration,      -41.2509433511
Predicted improvement:          0.001023175
lambda =          1; f =          -41.2528583
lambda =      1.9332; f =          -41.2544076
lambda =      3.7372; f =          -41.2567533
lambda =      7.2247; f =          -41.2588697
Norm of dx    0.0083206
----
Improvement on iteration 22 =          0.007926316
-----
f at the beginning of new iteration,      -41.2588696675
Predicted improvement:          0.005730041
lambda =          1; f =          -41.2702370
lambda =      1.9332; f =          -41.2806792
lambda =      3.7372; f =          -41.3004105
lambda =      7.2247; f =          -41.3368392
lambda =     13.967; f =          -41.4007511
lambda =         27; f =          -41.4992159
lambda =     52.196; f =          -41.5899070
Norm of dx    0.015048
----
Improvement on iteration 23 =          0.331037355
-----
f at the beginning of new iteration,      -41.5899070224
Predicted improvement:          0.333323662
lambda =          1; f =          -42.2428474
lambda =      1.9332; f =          -42.8274348
lambda =      3.7372; f =          -43.8898484
lambda =      7.2247; f =          -45.6892222
lambda =     13.967; f =          -48.1668657

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lambda =      27; f =      -39.7115865
lambda =     18.18; f =     -48.8828212
Norm of dx    0.69357
----
Improvement on iteration 24 =      7.292914150
-----
f at the beginning of new iteration,      -48.8828211720
Correct for low angle: 0.00355293
Predicted improvement:      1.805898410
lambda =      1; f =     -49.6041653
lambda =     0.33333; f =     -49.7260336
lambda =     0.64439; f =     -49.9381038
Norm of dx    3.4138
----
Improvement on iteration 25 =      1.055282594
-----
f at the beginning of new iteration,      -49.9381037656
Correct for low angle: 0.0040421
Predicted improvement:      0.231342777
lambda =      1; f =     -50.0275479
lambda =     0.33333; f =     -50.0543619
lambda =     0.64439; f =     -50.0886464
Norm of dx    1.2363
----
Improvement on iteration 26 =      0.150542595
-----
f at the beginning of new iteration,      -50.0886463606
Predicted improvement:      0.067513641
lambda =      1; f =     -49.9342815
lambda =     0.33333; f =     -50.1015206
lambda =     0.11111; f =     -50.1000741
lambda =     0.2148; f =     -50.1042991
Norm of dx    0.083216
----
Improvement on iteration 27 =      0.015652777
-----
f at the beginning of new iteration,      -50.1042991380
Predicted improvement:      0.000475082
lambda =      1; f =     -50.1047362
Norm of dx    0.029948
----
Improvement on iteration 28 =      0.000437012
-----
f at the beginning of new iteration,      -50.1047361501
Predicted improvement:      0.000004641
lambda =      1; f =     -50.1047405
Norm of dx    0.0014388
----
Improvement on iteration 29 =      0.000004333
-----
f at the beginning of new iteration,      -50.1047404835
Predicted improvement:      0.000000012
lambda =      1; f =     -50.1047405
Norm of dx    1.288e-05
----
Improvement on iteration 30 =      0.000000013
improvement < crit termination

```

Final value of minus the log posterior (or likelihood):-50.104740

MODE CHECK

Fval obtained by the optimization routine: -50.104740

RESULTS FROM POSTERIOR ESTIMATION

parameters

	prior mean	mode	s.d.	prior	pstdev
gamma_q	0.0000	0.3998	3.8717	norm	5.0000
gamma_N	0.0000	0.6638	0.0880	norm	5.0000

standard deviation of shocks

	prior mean	mode	s.d.	prior	pstdev
ep_N	0.0100	0.0521	0.0049	inv	0.0050
ep_Y	0.0100	0.2532	0.0238	inv	0.0050

Log data density [Laplace approximation] is 43.517048.

Estimation::mcmc: Multiple chains mode.

Estimation::mcmc: Old metropolis.log file successfully erased!

Estimation::mcmc: Creation of a new metropolis.log file.

Estimation::mcmc: Searching for initial values...

Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!

Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat

Estimation::mcmc: Number of mh files: 1 per block.

Estimation::mcmc: Total number of generated files: 2.

Estimation::mcmc: Total number of iterations: 125000.

Estimation::mcmc: Current acceptance ratio per chain:

Chain 1: 45.78%

Chain 2: 45.672%

Estimation::mcmc: Total number of MH draws per chain: 125000.

Estimation::mcmc: Total number of generated MH files: 1.

Estimation::mcmc: I'll use mh-files 1 to 1.

Estimation::mcmc: In MH-file number 1 I'll start at line 25001.

Estimation::mcmc: Finally I keep 100000 draws per chain.

MCMC Inefficiency factors per block

Parameter	Block 1	Block 2
SE_ep_N	25.260	23.245
SE_ep_Y	22.412	24.479
gamma_q	18.280	19.106
gamma_N	27.982	29.412

Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 1.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_ep_N	0.054	0.005	0.529	0.890	
0.882	0.865				
SE_ep_Y	0.261	0.025	0.362	0.853	
0.839	0.795				
gamma_q	1.992	3.491	0.175	0.726	
0.739	0.737				
gamma_N	0.699	0.103	0.000	0.343	
0.285	0.293				

Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 2.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_ep_N	0.054	0.005	0.000	0.290	
0.263	0.180				
SE_ep_Y	0.260	0.025	0.000	0.400	
0.369	0.289				
gamma_q	1.920	3.473	0.008	0.508	
0.520	0.515				
gamma_N	0.697	0.102	0.000	0.365	
0.369	0.337				

Univariate convergence diagnostic, Brooks and Gelman (1998):

Parameter 1... Done!

Parameter 2... Done!

Parameter 3... Done!

Parameter 4... Done!

marginal density: I'm computing the posterior mean and covariance... Done!

marginal density: I'm computing the posterior log marginal density (modified harmonic mean)... Done!

ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 43.530869.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
gamma_q	0.000	1.9671	-3.8066	7.6261	norm	5.0000
gamma_N	0.000	0.6982	0.5359	0.8589	norm	5.0000

standard deviation of shocks

	prior mean	post. mean	90% HPD interval		prior	pstdev
ep_N	0.010	0.0536	0.0452	0.0619	invga	0.0050
ep_Y	0.010	0.2604	0.2193	0.3004	invga	0.0050

```
--- Parameters Post. Std ---  
gamma_q: 3.481138  
gamma_N: 0.102014  
--- Shocks Post. Std ---  
ep_N: 0.005235  
ep_Y: 0.025335  
Total computing time : 0h13m46s  
>>
```